Competing Technology Adoption in Networked Systems

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digital futures

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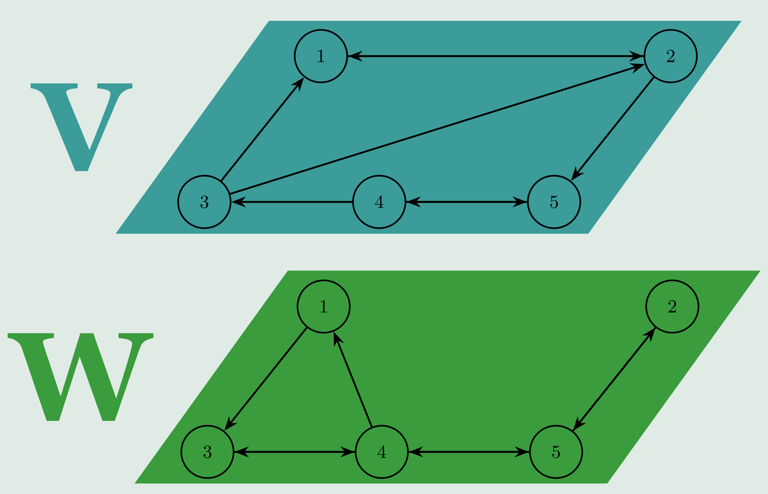
How does the network affect the diffusion of innovation?

The most significant impact is on the spreading timescale. Market dominance is also affected.

In this project, we evaluate a two-layer adoption-opinion model of multi-technology diffusion in networked systems. Specifically, we investigate empirically how network topology and connectivity affect the diffusion of technologies.

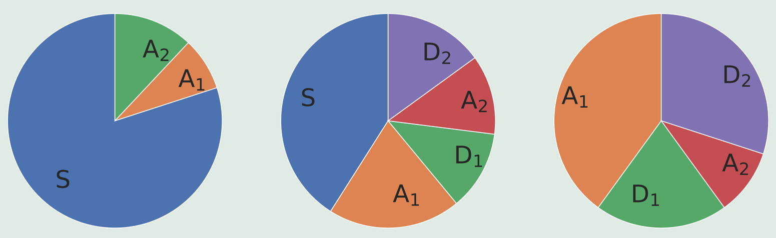
Bilayer model



The model represents a closed population of individuals, divided into n communities. The figure to the left shows a system with five communities. Communities are interconnected via two networks: a physical network (W) and a virtual one (V). Community 2, for example, is physically connected to community 5, while virtually connected to communities 1,3, and 5.

Susceptible, adopter, and dissatisfied

We model individuals as either susceptible to all technologies or as adopters/dissatisfied with a given technology; i.e., individuals are in one of 1 + 2K states, where K is the number of technologies. Individuals are influenced to change state by their own communities, by the adoption behavior of neighboring communities, and by their own experience with the technologies.
The figure below illustrates how ratios can change within a community.

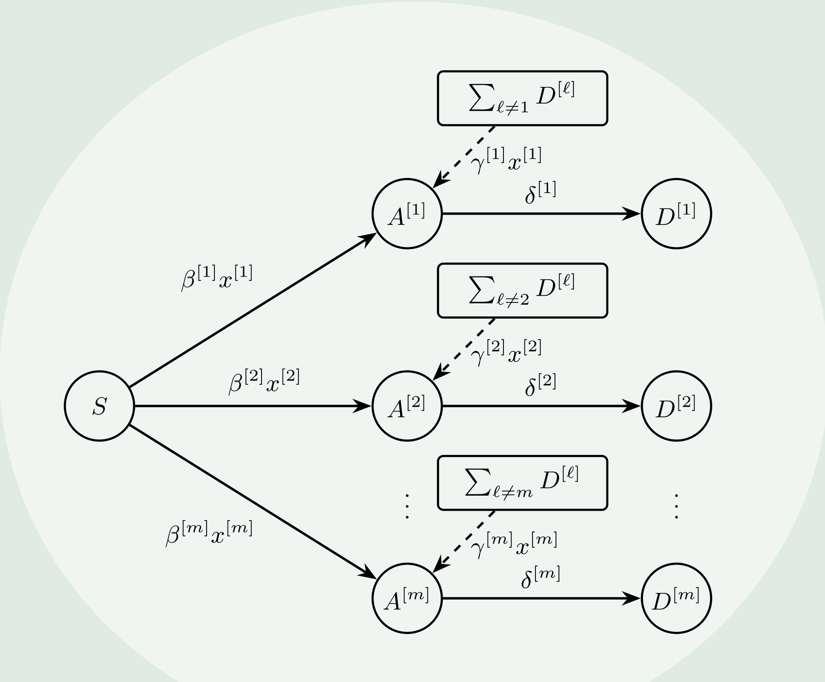


Naturally, the sum of all ratios adds up to one at all times.

$$s_i + \sum_{k\in K} a_i^{[k]} + d_i^{[k]} = 1$$

Opinion dynamics

The adoption of technologies also depends on the communities' opinions of the technologies. Each community has an opinion on each technology; the dynamics of a given community's opinions are affected by the adoption and opinions of neighboring communities, as well as its initial opinion and its degree of stubbornness.



The coupled model

Coupling the adoption process with opinion dynamics gives the following model.

$$s_i(t+1) = s_i(t) - s_i(t)\sum_{k\in K}\beta_i^{[k]}x_i^{[k]}(t)\sum_{j\in n}W_{ij}a_j^{[k]}(t)$$

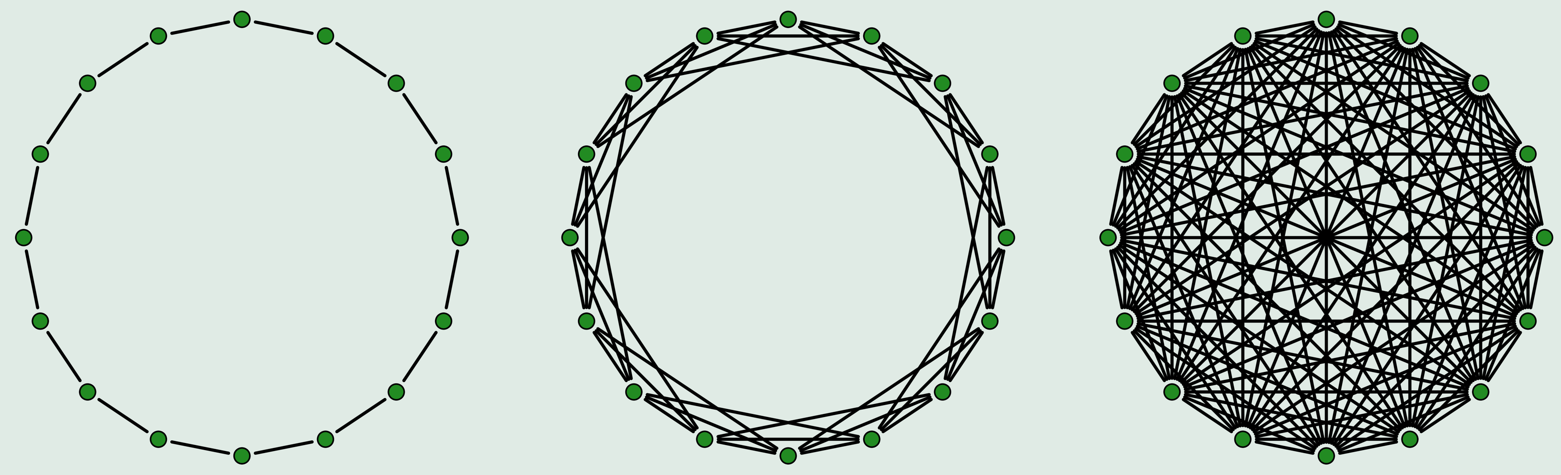
$$a_i^{[k]}(t+1) = a_i^{[k]}(t) + s_i(t)\beta_i^{[k]}x_i^{[k]}(t)\sum_{j\in n}W_{ij}a_j^{[k]}(t)$$

$$+\gamma_i^{[k]}x_i^{[k]}(t)\sum_{l\neq k}d_i^{[l]}(t) - \delta_i^{[k]}a_i^{[k]}(t)$$

$$d_i^{[k]}(t+1) = d_i^{[k]}(t) - d_i^{[k]}(t)\sum_{l\neq k}\gamma_i^{[l]}x_i^{[l]}(t) + \delta_i^{[k]}a_i^{[k]}(t)$$

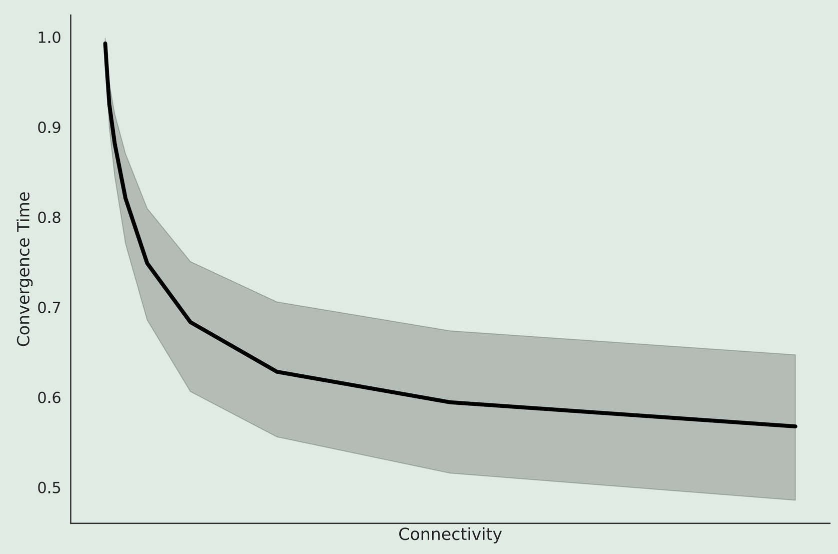
$$x_i^{[k]}(t+1) = (1-\lambda_i^{[k]}-\xi_i^{[k]})x_i^{[k]}(0) + \lambda_i^{[k]}\sum_{j\in n}V_{ij}x_j(t) + \xi_i^{[k]}\sum_{j\in n}W_{ij}a_j^{[k]}(t)$$

Experiments



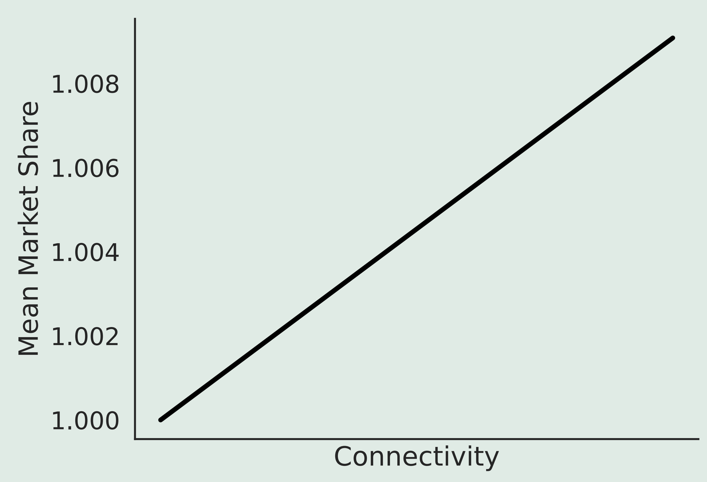
Convergence time and connectivity

We measured how the level of connectivity in W affects the convergence time to the steady state, while keeping V as a random complete graph at all times. The vertical axis is the normalized convergence time; the horizontal axis is the number of directly connected neighbors.
The middle line is the mean of all experiments; the darker area is the 95% confidence interval.

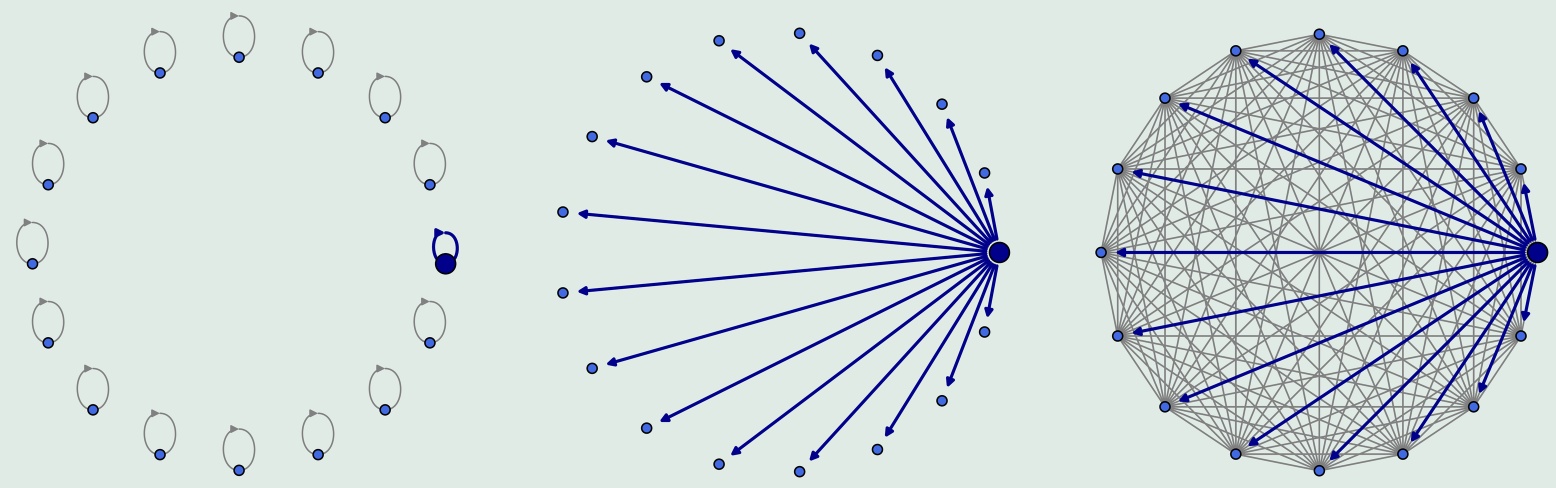


Market share dominance and connectivity

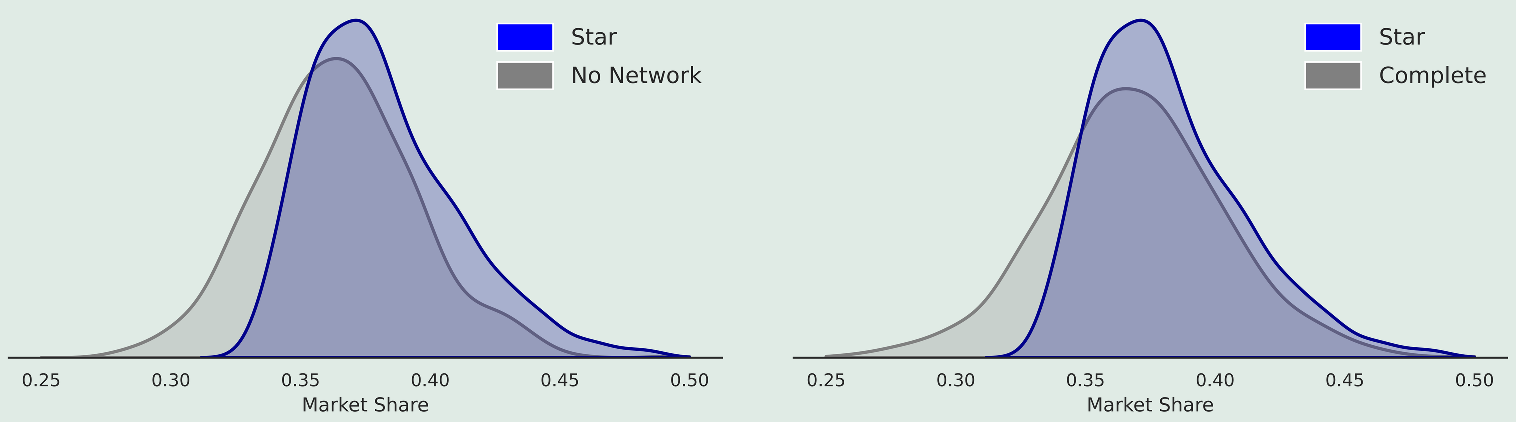
In the same experiment, we measured the difference in market shares of the dominant technology in the ring and in the complete graph.
The mean increase across all experiments is plotted on the right; the difference between the least- and most-connected networks is less than 1%.



Social media influencers and market share

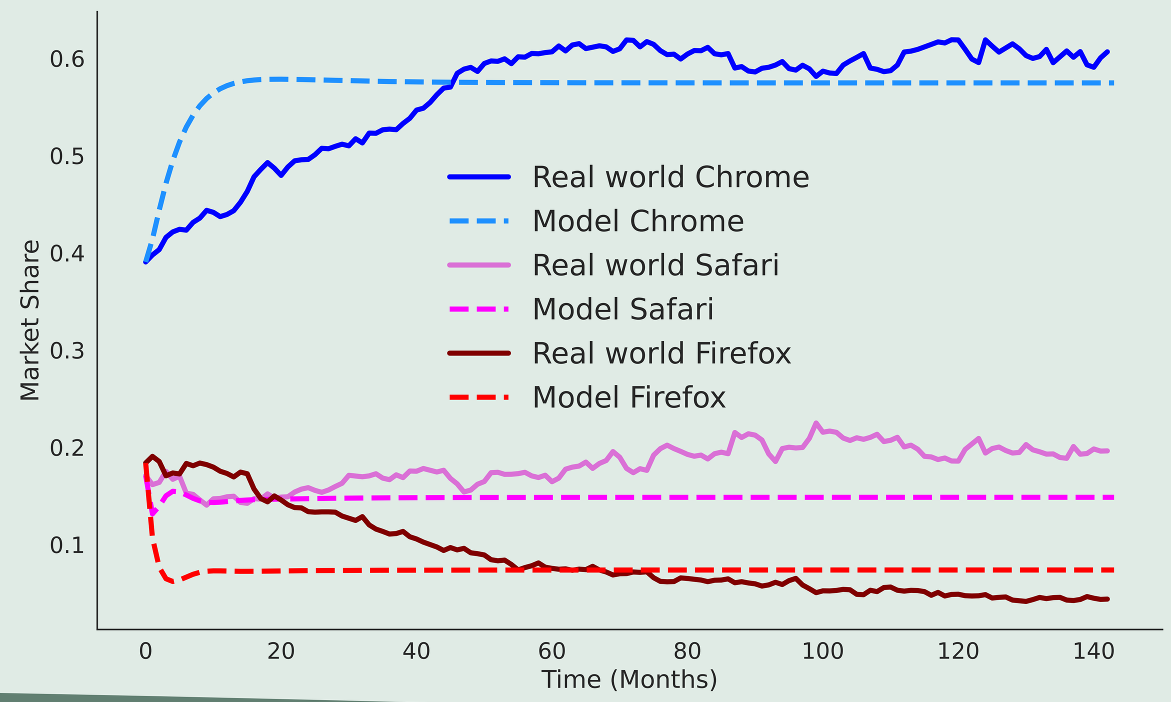


We measured how technology influencers affect the final market share distribution of a promoted technology. An influencer community has a high level of initial adoption and a strong, positive opinion of the given technology, while also being stubborn.
While keeping all other parameters fixed, we analyze how V's topology affects the market share of a selected technology.



Model validation

We validated the model against real world data. The figure below shows the top three web browsers in Europe from January 2015 to January 2026 and the model's result.



References: Alutto et al. "Predictive control strategies for sustaining innovation adoption on multilayer social networks", arXiv preprint arXiv:2509.01457, 2025.
Alutto et al. "On a Coupled Adoption–Opinion Framework for Competing Innovations”, Proceedings of the 23rd IFAC World Congress, 2026.